

Assignment #2 Report

Course: EE 8310 – Advanced Topics in VLSI

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Problem 1: Effect of 4th Inverter Load

1.1 Objective

In class, we discussed the effect of adding the “4th inverter load” when characterizing the inverter delay. In this assignment, compare the signals waveforms and delays of an inverter chain with and without the 4th inverter and check if the delay of the 2nd inverter indeed changes between the two configurations. Draw the schematic (no layout required) in TSMC N16 academic PDK using an inverter chain with 4x size standard-Vt inverters. Simulations should be performed under nominal VDD conditions at a temperature of 80°C.

1.2 Methodology

- Built two inverter chains using 4× sized inverters.
- Case 1: 3 inverters (no extra load).
- Case 2: 4 inverters (extra load).
- Input: step input at Inverter 1.
- Output delays measured at Inverter 2 in both cases.

1.3 Simulation Setup

- Technology: TSMC N16, SVT, 4× inverters.
- VDD = 0.8 V, Temp = 80 °C.
- Transient analysis with input pulse source.

1.4 Results

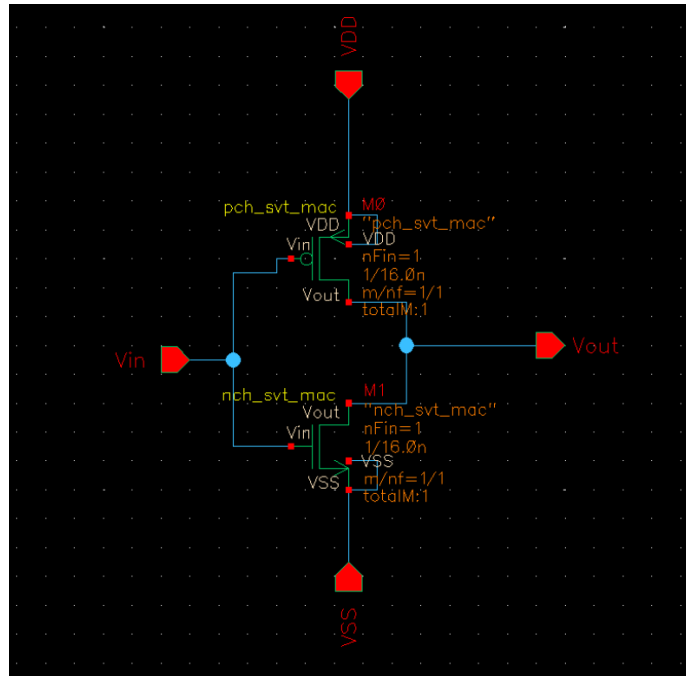


Figure : Inverter

3 Inverters

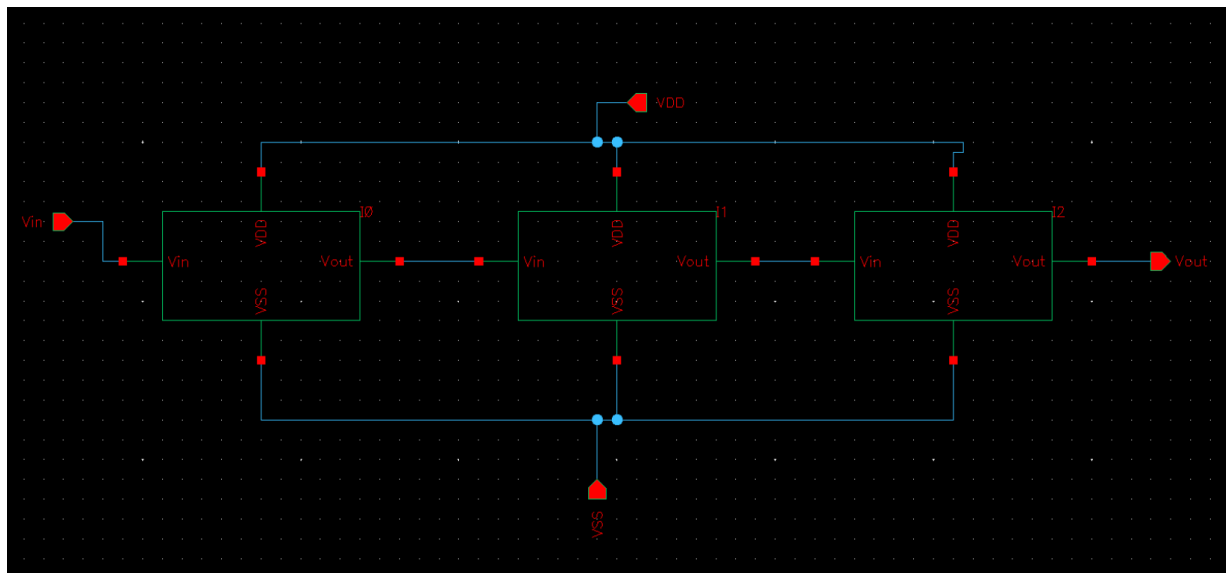


Figure: 3 Stage inverter chain

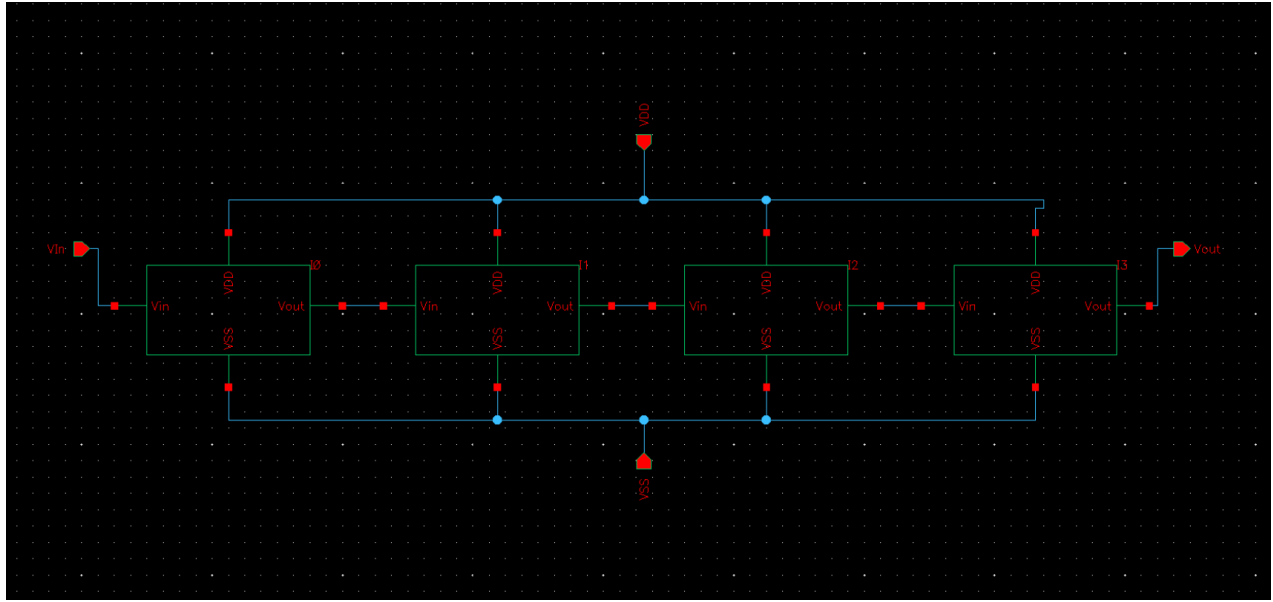


Figure: 4 Stage inverter chain

Simulation testbench was same for both the inverters and the delay values were calculated.

```
.TEMP 80
.OPTION POST
.lib
/soft/ece_soft_2/TSMCHOME/Executable_Package/Collaterals/Tech/SPICE/N1
6ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0_usage.1 TTMacro_MOS_MOSCAP

.INCLUDE inv4.sp

* Voltage sources
Vin Vin 0 PULSE(0 0.8V 60p 10p 10p 60p 140p)
Vdd VDD 0 0.8V
Vss VSS 0 0V

* Small 1fF output capacitance for realistic simulation
C1 Y 0 2fF
.OP
.TRAN 0.1p 1n

*.MEASURE TRAN th1 TRIG V(a) VAL=0.35 TD=100p RISE=2 TARG V(y)
VAL=0.35 TD=100p FALL=2
*.measure tran freq param= '20/t1'
*.measure power rms power
.END
```

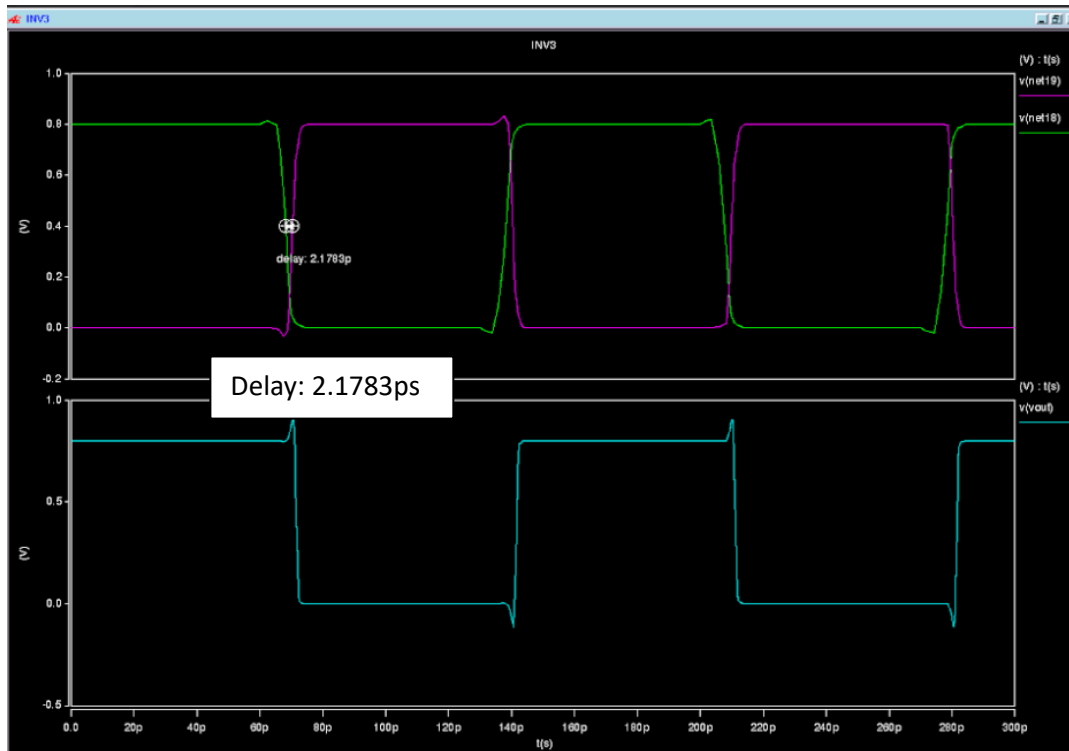


Figure: Delay of Inverter 2 in the 3 inverter chain

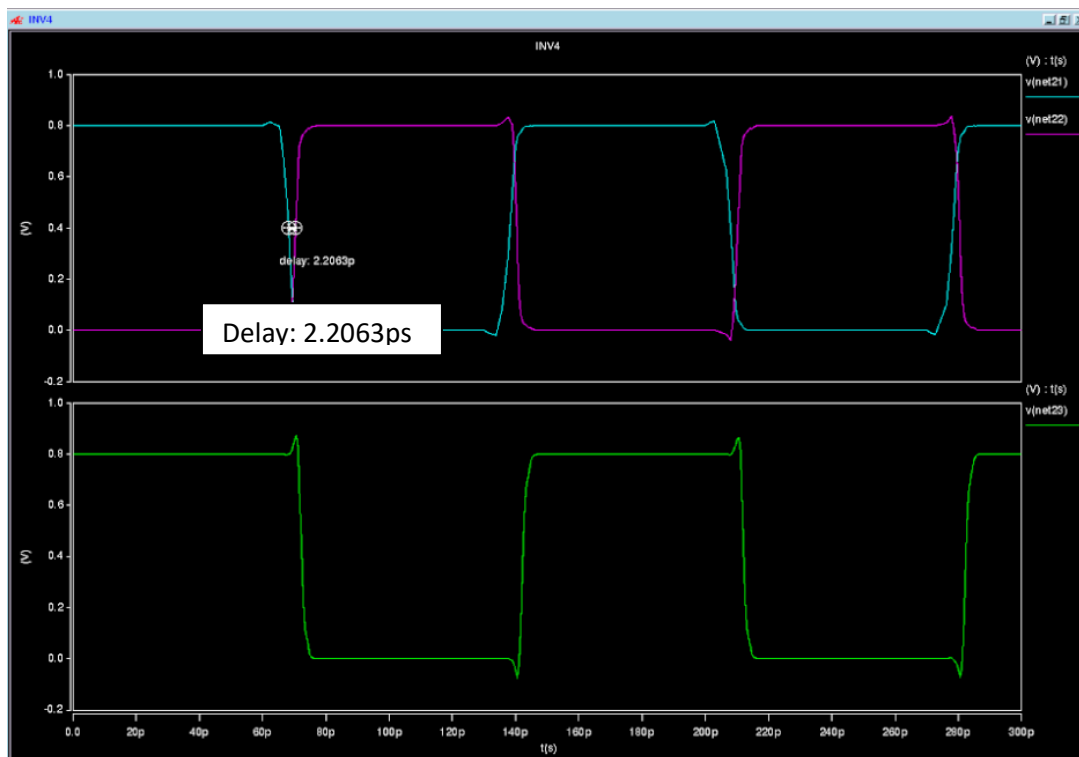


Figure: Delay of Inverter 2 in the 4 inverter chain

Overshoot and undershoot from VDD and VSS in respective cases.

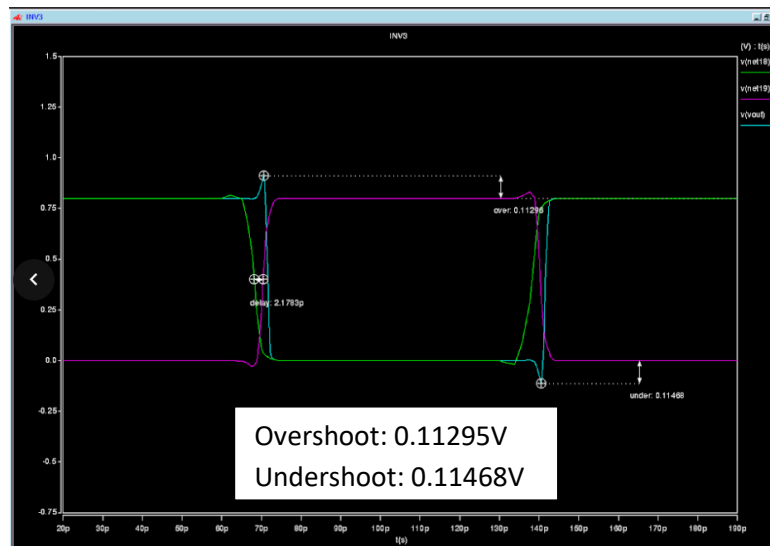


Figure: INV3

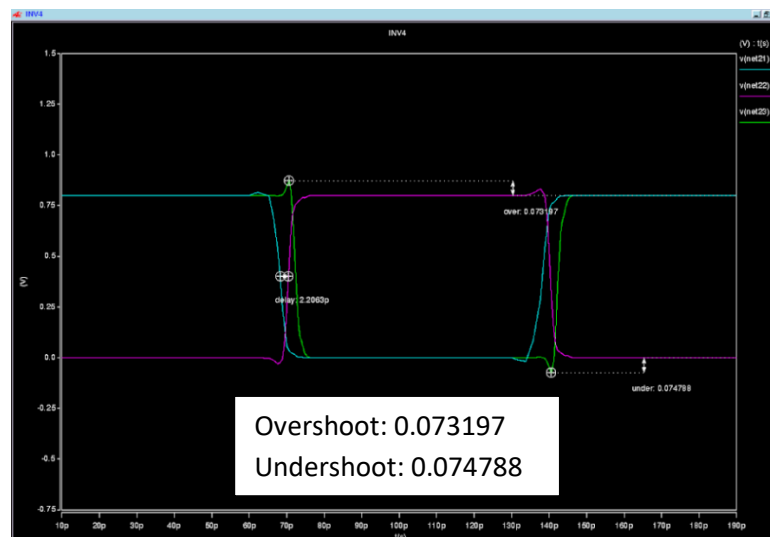


Figure: INV4

Analysis

Transistor Miller (C_{gd}) feedback in INV3 and slew-dependent coupling reflect part of INV3's output load back to its input, making INV2 see a larger effective capacitance when the 4th inverter is added. This matches simulation: INV2 delay is smaller in the 3-inverter chain and increases slightly when the 4th inverter load is present.

Problem 2: 15-Stage Ring Oscillators with Different Sizes

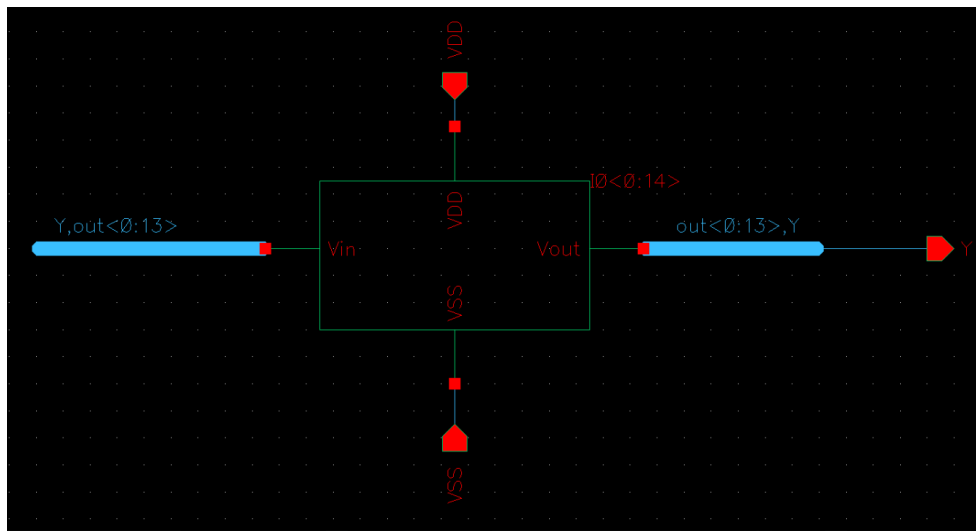
2.1 Objective

To observe frequency scaling with inverter size ($1\times$, $2\times$, $4\times$, $8\times$).

2.2 Methodology

- Designed four 15-stage schematics.
- Extracted oscillation frequency via transient simulation.

2.3 Schematics and Setup



2.4 Simulation Testbench and Results

```
.TEMP 80
.OPTION POST
.lib
/soft/ece_soft_2/TSMCHOME/Executable_Package/Collaterals/Tech/SPICE/N1
6ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0_usage.1 TTMacro_MOS_MOSCAP

.INCLUDE ringOscilx.sp

* Voltage sources
Vy Y 0 PULSE(0 0.8V 100p 10p 10p 80p 180p)
Vdd VDD 0 0.8V
Vss VSS 0 0V

* Small 1fF output capacitance for realistic simulation
C1 Y 0 2fF
.OP
.TRAN 0.1p 2n
```

```

.MEASURE TRAN tper TRIG V(Y) VAL=0.4 RISE=1 TARG V(Y) VAL=0.4 RISE=2
.MEASURE TRAN freq PARAM='1/tper'
.MEASURE tran power avg 'frequency*C1*((VDD)^2)'
.MEASURE tran iavg avg I(V2)
.MEASURE tran power param='V(VDD)*iavg'
.MEASURE tran power P(VDD)
.MEASURE tran power avg POWER

```

Inverter Size	Frequency Trend	Max Frequency	Min Frequency
1x	5.5562 GHz 5.5559 GHz 5.5524 GHz	5.5562 GHz	5.5524 GHz
2x	5.5567 GHz 5.555 GHz 5.5559 GHz	5.5576 GHz	5.5532 GHz
4x	5.5567 GHz 5.5561 GHz 5.5556 GHz	5.5575 GHz	5.554 GHz
8x	5.5542 GHz 5.5548 GHz 5.5547 GHz	5.5589 GHz	5.5521 GHz

2.5 Analysis and Observations

The frequencies for 1×, 2×, 4×, and 8× inverters were nearly identical (~5.55 GHz), confirming the theoretical expectation that increasing inverter size does not significantly affect ring oscillator frequency. This is because the reduced delay from stronger drive is offset by the increased input capacitance of subsequent stages. Minor variations (within ~0.1%) are attributed to parasitic capacitances and simulation tolerances.

Problem 3: 101-Stage Ring Oscillator (Bus Notation)

4.1 Objective

Implement the schematic of a 101 stage ring oscillator using a single inverter symbol and the bus notation method. Simulate the frequency under nominal VDD and a temperature of 80°C.

4.2 Schematic and Implementation

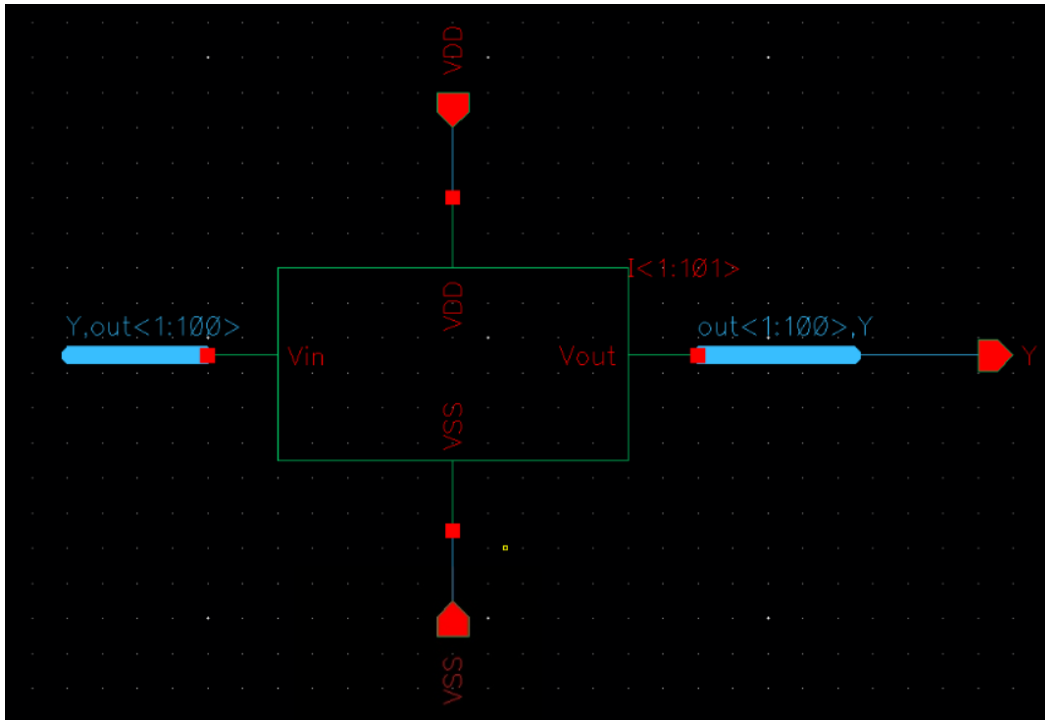


Figure: Schematic of a 101 Stage Ring Oscillator using bus notation

4.3 Simulation Results

